



TAYLOR LANE

Crestview, FL 32536

taylor.a.lane.civ@gmail.com / (936)-524-9786

PROFESSIONAL SUMMARY

Motivated mechanical engineer with experience in system design, equipment validation, and performance optimization. Skilled in troubleshooting, diagnostics, and improving performance of mechanical and electromechanical systems in high-demand environments. Strong foundation in thermodynamics, data analysis, and engineering problem-solving, with a focus on delivering efficient and reliable solutions.

SKILLS

- Systems engineering
- Python
- Mechanical design
- LabVIEW, MATLAB
- Material testing techniques
- SolidWorks

EXPERIENCE

Special Forces Communication Sergeant / United States Army - Eglin Airforce Base, FL *05/2021 - Current*

- Established robust communications plans for a Special Forces Company that ensured seamless connectivity across the unit
- Coordinated geographically dispersed teams to execute time-sensitive communications protocols, ensuring continuous operations
- Diagnosed and repaired advanced communications systems under extreme pressure, rapidly restoring full functionality and minimizing operational downtime
- Led pre-mission equipment diagnostics and functionality testing to reduce risk of in-operation failures
- Implemented communications risk matrix to assess vulnerabilities, and inform mission planning

Design Engineer / Goodman Manufacturing - Waller, Texas *02/2021 - 05/2021*

- Analyzed extensive BOM datasets to support risk mitigation and inform design decisions that improved safety.
- Developed an Excel VBA program to parse outdoor unit BOMs, identifying fire-risk components to enhance safety measures.
- Collaborated with a diverse engineering team to implement innovative solutions, improving project outcomes.

Mechanical Engineering Intern / Trane - Clarksville, TN *05/2019 - 08/2019*

- Forecasted HVAC fan configurations requiring modifications for compliance with Title 20 Fan Energy Index (FEI) efficiency regulations.
- Established a standard torque testing procedure for new HVAC panel insulation products, obtaining baseline performance data.
- Collaborated with engineers to develop a parsing tool that generated CAD sketches based on product number inputs.
- Identified HVAC louvered panel design reductions, verified through airflow testing, to decrease production and operation costs.

Systems Engineer Intern / Trane - Tyler, TX

08/2018 - 12/2018

- Implemented system simulation software, enabling customers to achieve optimal HVAC systems that met performance specifications.
 - Conducted testing on prototype flow meter designs, enhancing troubleshooting methods for residential HVAC systems.
 - Collaborated with HVAC subject matter experts and marketing teams to assess potential for regional utility rebates on new systems.
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EDUCATION

Bachelor's of Science: Mechanical Engineering
Texas A&M University - College Station, TX
GPA: 3.4/4.0

08/2020